SOVEREIGN BRIEFING

First Receipted Human-in-the-Loop Critical Resolution in an Autonomous Self-Healing Infrastructure

Prepared for sovereign infrastructure, defense, and regulatory audiences · Event window: 2026-10-04, 23:10–23:18 CT · Evidence of record: public append-only ledger (github.com/LLong2026/four-horsemen, entries 0009–0010, commits 700ea1e, c1f43bb, ccf691a) · Zenodo DOI 10.5281/zenodo.23151356 · Patent cross-reference: 64/157,915 (filed 2026-09-18)

1. What Happened

On October 4, 2026, at 23:10 CT, an autonomous self-healing infrastructure (SQUIRL OS) detected a critical-severity anomaly during its nightly chaos-and-learning training cycle: a post-quantum cryptographic (PQC) keyring desynchronization, detected at confidence 0.91 against a 0.88 playbook threshold.

The system refused to heal it.

Because the anomaly was critical-severity and in a cryptographic flow, the governance layer held the event for human release. For eight minutes the anomaly sat gated — detected, contained, escalated, and deliberately unremediated. At 23:18 CT the human operator of record issued a formal acknowledgment. The guarded heal then executed in 22 seconds via the PQC Keyring Desync Recovery playbook, and the complete trail — refusal, hold, human release, execution, verification — was committed to the public append-only ledger within minutes of the event, and archived under DOI 10.5281/zenodo.23151356.

Full gated chain: detect → hold → constitutional check → PQC validation → human ACK → heal (22s) → resolve

2. Why It Matters

The standing objection to autonomous remediation in sovereign infrastructure is not capability — it is accountability: *if the machine can act, who answers for it, and can oversight be proven after the fact?*

Industry approval workflows exist across AIOps platforms, but they produce private, mutable logs. They assert oversight; they do not prove it. This event demonstrates the missing evidence shape: the oversight itself is receipted — publicly, append-only, timestamped, third-party verifiable from commit history alone. A regulator, insurer, or auditor can independently confirm: (1) the system declined to act on a critical event without a human — the refusal is a recorded artifact, not an absence; (2) a named human released the action at a specific time; (3) the action that followed is recorded end-to-end, with immutable provenance. For sovereign adoption of autonomous systems in critical infrastructure, this is the certification pattern: **not trust — receipts.**

3. Substrate Families Involved

Substrate A — Deterministic runtime. The purely deterministic execution layer, previously certified by SDR-1 (Substrate Determinism Receipt 1): identical outputs on identical inputs across instances and wall-clock times, byte-identical replay parity (checksum 0xc5f88dcf, 4/4 parity, 24/24 replay identity, temporal independence).

Substrate B — Bifurcated runtime. The probabilistic-proposer-under-deterministic-cage runtime in which this event occurred: a probabilistic agent proposed the detection (with confidence score), and the deterministic governance layer validated, gated, and executed. Substrate B previously demonstrated 100% pattern parity in independent benchmarking (September 29, 2026).

The event's control plane ran through the SQUIRL OS Hub (mission control), with the target in a simulated client-hosted environment (SIMZ corpus). The deterministic layer — not the probabilistic layer — held the gate.

4. Governance Doctrine That Fired

The bifurcated governance doctrine: **probabilistic agents propose; a deterministic substrate validates against constitutional invariants before execution.** No probabilistic output executes without passing the deterministic gate. Instruments engaged in this event, in order:

#	Instrument	Effect in this event
1	Human-in-the-loop gate (critical cryptographic flows)	Produced the refusal — critical-severity crypto anomalies are never auto-healed
2	Constitutional audit (Articles I–III, 12-article Constitution)	Deterministic protocol check passed before the critical alert was created — audit is code, not model judgment
3	Playbook immutability + exact-match enforcement	Remediation only via a playbook whose anomaly type exactly matches
4	Confidence thresholding	0.91 detection confidence checked against the 0.88 playbook threshold before any action
5	Full-context audit mandate	No healing without a logged event carrying anomaly, playbook, steps, result, duration, agent, node
6	Redaction at ingestion	No PII, credentials, wallet addresses, or transaction amounts persisted — structural/operational metadata only

The constitutional audit layer is a deterministic protocol — not a probabilistic agent — so it cannot negotiate its own rules. Governance rules are human-amendable only.

5. PQC Lineage Enforced

The cryptographic doctrine is **PQC-only**, enforced at validation time rather than by convention:
• **Algorithm used:** CRYSTALS-Dilithium3 (approved lineage: CRYSTALS-Dilithium3 / Kyber-1024 / SPHINCS+-256f). The keyring rotation executed exclusively through the approved PQC path.
• **Non-PQC fallback rejected:** the validation step confirmed no classical or unapproved cryptographic fallback exists in the remediation path — fallback is prohibited, not merely discouraged.
• **Pre-heal cryptographic validation:** a dedicated check ran before the healing action touched key rotation, per the standing rule that post-quantum compliance is validated prior to any cryptographic operation.
• **Crypto-agility schema:** anomalies carry quantum-threat metadata (threat level, vulnerable algorithm, suggested PQC algorithm, estimated break year), per the PQC Lineage Map v1.2 (DOI 10.5281/zenodo.22863667).

The significance for sovereign cryptographic post-quantum migration programs: the same governance gate that held for a human answer also enforced the post-quantum lineage. Migration compliance and human accountability were enforced by one deterministic instrument.

6. Why This Is a First

To our knowledge, no prior public record exists of an autonomous self-healing infrastructure **receipting its own refusal** to act on a critical event, a **named human's release**, and the **resulting execution** as one continuous, publicly verifiable, append-only trail — committed within minutes of the event.
• Prior AIOps approval workflows are private and mutable; none produce a public, append-only receipt of the oversight itself.
• Prior human-in-the-loop frameworks (ML oversight, incident-response sign-offs, two-man rules) assert oversight after the fact; none make the oversight a deterministic artifact with immutable provenance.

• A prior acknowledgment event exists in this program's internal records (September 30, 2026) but was never publicly receipted; this cycle is the first committed end-to-end to a public ledger, and the distinction is itself disclosed — receipted evidence and unreceipted assertion are not the same class of claim.

The governing method is the subject of pending patent application 64/157,915 (Deterministic Constitutional Autonomous Infrastructure Systems and Methods). The mathematical companion is the Governed Manifold Specification v1.1 (DOI 10.5281/zenodo.23132046).

Disclosures

The anomaly was a synthetic chaos-injection probe (Night 10 of a nightly training cycle) on a simulated target (SIMZ corpus); no real tenant, customer, or live cryptographic operation was involved. Night 10 executed as a disclosed manual recovery: the scheduled workflow fired on time but its spawned session lacked cross-app write access (platform limitation, disclosed in ledger entry 0009); the recovery session performed the cycle under a fresh idempotency stamp. Prior nights N7–N9 (October 1–3) are recorded as missed, not backfilled. Every claim above is checkable against the public ledger and archived record.

Record Index

Artifact	Location
H-Line ledger entries 0009–0010	github.com/LLong2026/four-horsemen/LEDGER.md
N10-ACK milestone receipt	MILESTONE_HITL_N10_Critical_Receipt.md, same repository
Archived achievement paper	DOI 10.5281/zenodo.23151356
Governed Manifold Specification v1.1	DOI 10.5281/zenodo.23132046
PQC Lineage Map v1.2	DOI 10.5281/zenodo.22863667
Governing patent application	64/157,915 (filed 2026-09-18)

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